Osptek Display

SPECIFICATION FOR 4 . 2 "EPD

Model No:

EPD042A02A0

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Version	Content	Date	Producer
A0	New release	2026/10/22	HT

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1. General Description

EPD042A02A0 is an Active Matrix Electrophoretic Display (AMEPD), with interface and a reference system design. The 4.2" active area contains 300×400 pixels, and has 3-bit B/W/R/Y/B/G full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

2. Features

- 300×400 pixels display
- High contrast

  High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Low voltage detect for supply voltage
- High voltage ready detect for driving voltage
- Internal temperature sensor
- Waveform stored in On-chip OTP
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- I2C signal master interface to read external temperature sensor/ built-in temperature sensor

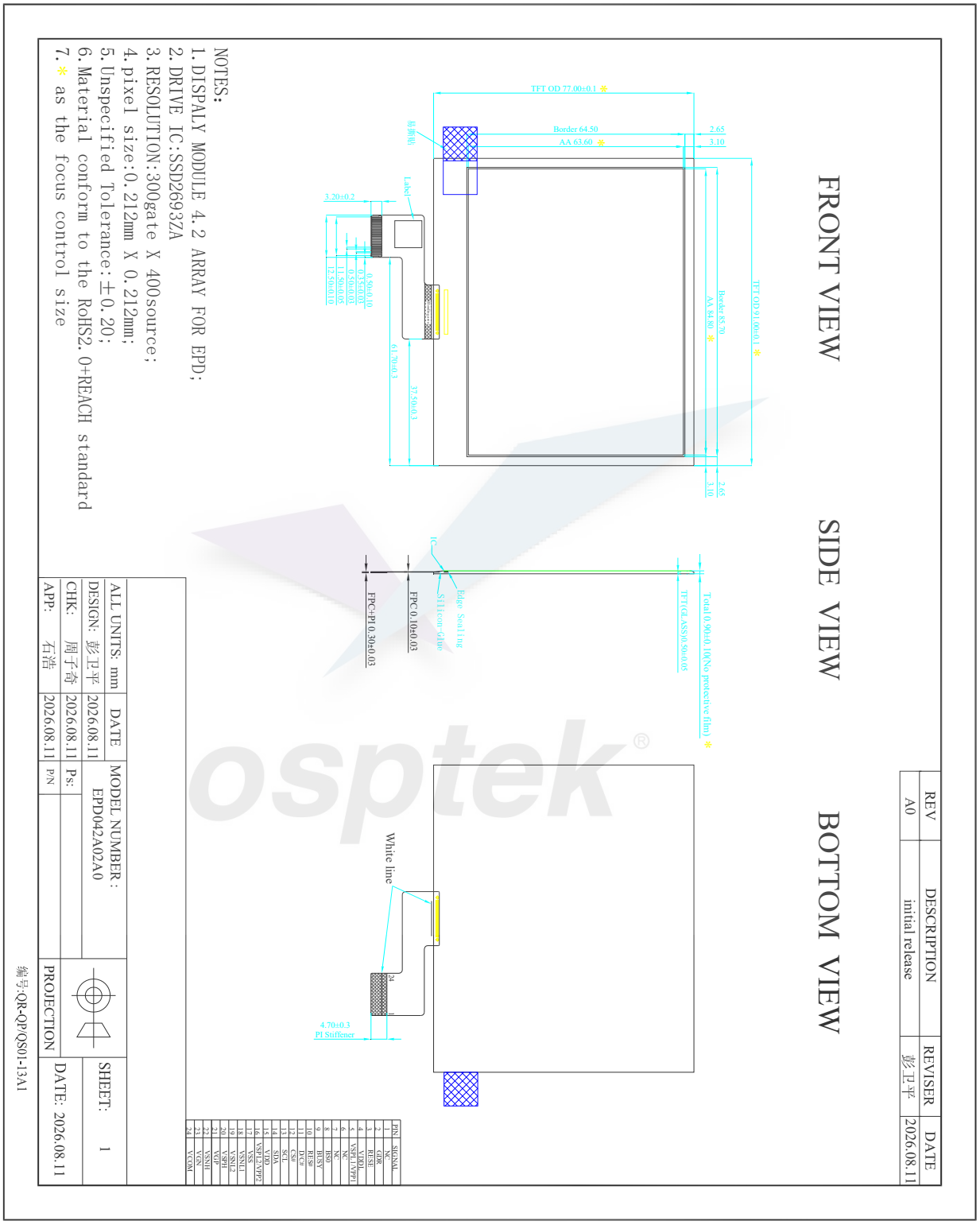
3. Application

Electronic Shelf Label System

4. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	4.2	Inch	
Display Resolution	400(H)×300(V)	Pixel	Dpi:119
Active Area	84.8(H)×63.6 (V)	mm	
Pixel Pitch	0.212×0.212	mm	
Pixel Configuration	Rectangle		
Outline Dimension	91.00(H)× 77.00(V) ×0.9(D)	mm	Without masking film
Weight	15±0.5	g	

5. Mechanical Drawing of EPD module



PIN	SIGNAL
1	NC
2	GDR
3	RESE
4	VSPL1/VPP1
5	VSPH1/VDD1
6	NC
7	BS0
8	BUSY
9	RES#
10	D/C#
11	SCL
12	SDA
13	VDD
14	VSPL2/VPP2
15	VSPH2/VDD2
16	VSSL1
17	VSSH1
18	VSPL
19	VSPH
20	VGH
21	VSSH
22	VGN
23	VCOM

6. Input/output Terminals

Pin #	Single	Description	Remark
1	NC	No connection and do not connect with other NC pins	Keep Open
2	GDR	N-Channel MOSFET Gate Drive Control	
3	RESE	Current Sense Input for the Control Loop	
4	VDDL	No connection and do not connect with other NC pins	Keep Open
5	VSPL1/VPP1	Positive Source driving voltage	
6	NC		Note 6-6
7	NC		Note 6-6
8	BS0	Bus selection pin	Note 6-5
9	BUSY	Busy state output pin	Note 6-4
10	RES #	Reset signal input.	Note 6-3
11	D/C#	Data /Command control pin	Note 6-2
12	CS#	The chip select input connecting to the MCU.	Note 6-1
13	SCL	Serial clock pin for interface.	
14	SDA	Serial data pin for interface.	
15	VDD	Power input pin for the Interface.	
16	VSPL2/VPP2	Power Supply pin for the chip	
17	VSS	Ground (Digital)	
18	VSNL1	Core logic power pin	
19	VSNL2	Power Supply for OTP Programming	
20	VSPH	Positive Source driving voltage	
21	VGP	Power Supply pin for Positive Gate driving voltage	
22	VSNH	Negative Source driving voltage	
23	VGN	Power Supply pin for Negative Gate driving voltage	
24	VCOM	VCOM driving voltage	

Note 6-1: This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication: only when CS# is pulled LOW.

Note 6-2: This pin (D/C#) is Data/Command control pin connecting to the MCU.

When the pin is pulled HIGH, the data will be interpreted as data. When the pin is

pulled LOW, the data will be interpreted as command.

Note 6-3: This pin (RES#) is reset signal input. The Reset is active low.

Note 6-4: This pin (BUSY) is Busy state output pin. When Busy is Low ,the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin Low when the driver IC is working such as:

- Outputting display waveform; or
- Communicating with digital temperature sensor

Note 6-5: This pin (BS1) is for 3-line SPI or 4-line SPI selection. When it is "Low", 4-line SPI is selected. When it is "High", 3-line SPI (9 bits SPI) is selected.

Note 6-6：We had made Pin 6 and Pin 7 on FPC to be ground.

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7. Reference Circuit

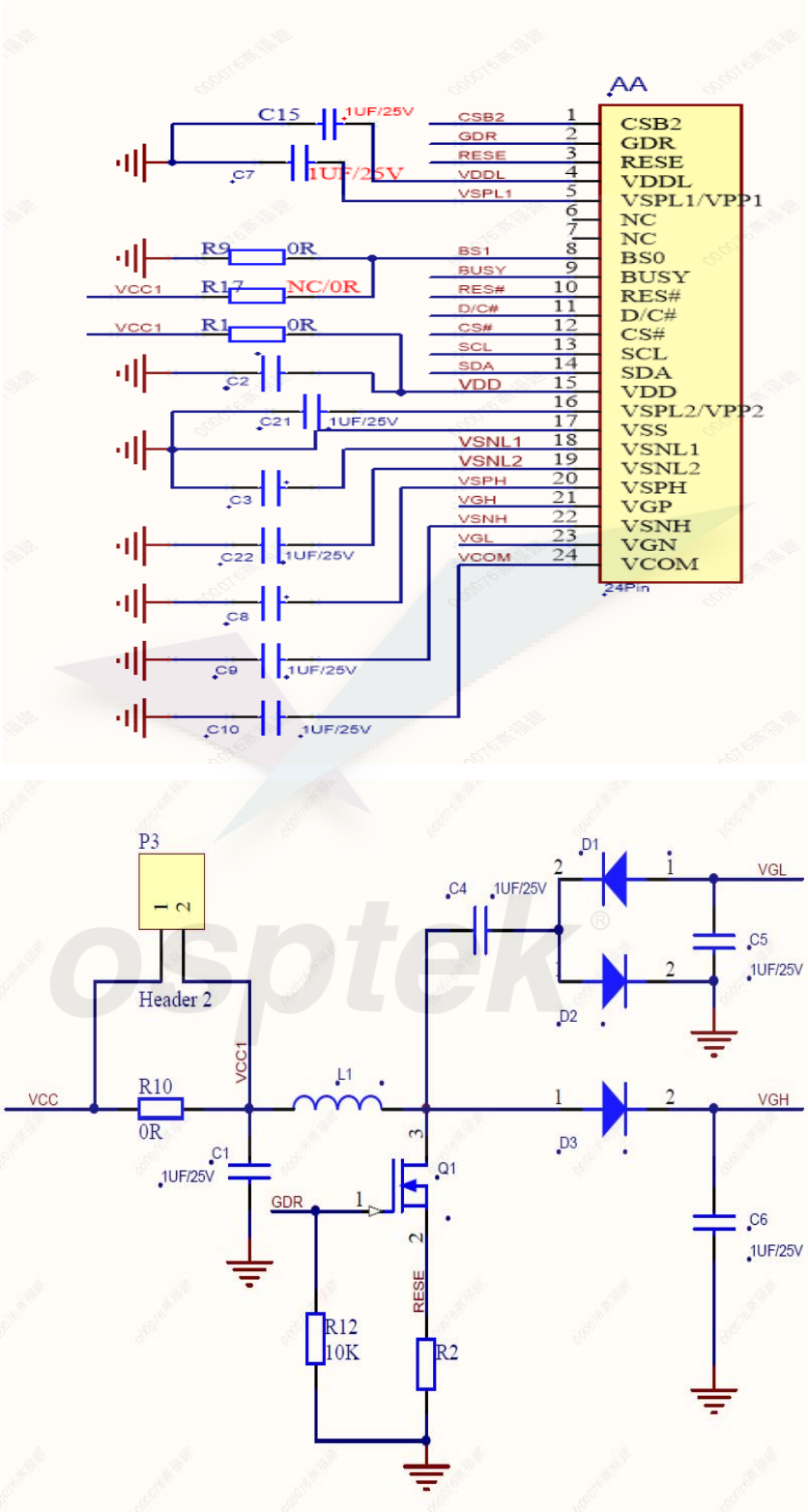


Figure. 7-1

Part Name	SSD2693ZAValue /requirement/Reference Part
C1-C4	1uF/0603;X5R;Voltage Rating: 25V
C5-C6	1uF/0603;X5R;Voltage Rating: 25V
C7-C10	1uF/0603;X5R;Voltage Rating: 25V
C15	1uF/0603;X5R;Voltage Rating: 25V
C21-C22	1uF/0603;X5R;Voltage Rating: 25V
D1—D3	MBR0530 1) Reverse DC voltage≥30V 2) Forward current≥500mA 3)Forward voltage≤430mV
R2	2.2 Ω/0603：1% variation
Q1	NMOS:Si1304BDL/NX3008NBK 1） Drain-Source breakdown voltage ≥30V 2） Vgs（th）=0.9（Typ），1.3V（Max） 3） Rds on≤ 2.1Ω @ Vgs=2.5V
L1	47uH: NRH3010T470MN

Figure. 7-2

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8. ABSOLUTE MAXIMUM RATING

Symbol	Parameter	Rating	Unit	Humidity	Unit	Note
V_{DD}	Logic supply voltage	-0.3 to +6.0	V	-	-	
T_{OPR}	Operation temperature range	0 to 50	℃	45 to70	%	
Tttg	Transportation temperature range	-25 to 50	℃	45 to70	-	Note11-2
Tstg	Storage condition	-25 to 50	℃	45 to70	%	Maximum storage time: 5 years

Note 8-1:We guarantee the single pixel display quality for 0-35℃, but we only guarantee the barcode readable for 35-40℃.Normal use is recommended to refresh every 24 hours.

Note 8-2:Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Electrical Characteristics chapter.

Not e8-3: Tttg is the transportation condition, the transport time is within 10 days for -25℃~0℃ or 50℃~60℃

Note 8-4： When the six-color product is stored. The display screen should be kept white and face up.We suggest that the full black and full white picture could be added to clear the screen after the module is refreshed for a long time, the display effect would be better.

Note 8-5： Due to the nature of FPL, we will not control the display effect of yellow and white characters on white and yellow characters on white.

9. DC Characteristics

The following specifications apply for: VSS=0V, VDD=3.0V, TOPR=25℃

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
VDD	VDD operation voltage	-	2. 4	3.0	3.6	V
VIH	High level input voltage	-	0.7xVIO	-	VIO	V
VIL	Low level input voltage	-	GND	-	0.3xVDD	V
VOH	High level output voltage	IOH = 400μA	VIO-0.4	-	-	V
VOL	Low level output voltage	IOL = -400μA	GND	-	GND+0.4	V
Iupdate	Module operating current	-	-	TBD	-	mA
Isleep	Deep sleep mode	VCI=3.3V	-	TBD		uA

- The Typical power consumption is measured using associated 25℃ waveform with followingpattern transition: from horizontal scan pattern to vertical scan pattern.
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by XingTai.

10. Power Consumption

Parameter	Symbol	Conditions	TYP	Max	Unit	Remark
Panel power consumption during update	-	25℃	TBD		mAs	-
Deep sleep mode	-	25℃	TBD		uA	-

mAs=update average current×update time

11. Optical characteristics

11.1 Optical Measurement Conditions

Item	Symbol	Value	Unit	Note
Ambient Temperature	Ta	25±2	℃	Indoor testing
Ambient Humidity	Ha	50±5	%RH	-
Supply Voltage	VCC, VDDIO	3.0	V	-
illuminance	-	800~1300	Lux	

Note 11-1: Image is updated with above condition

11.2. Optical characteristics

T=25℃ ,VDD=3.3V

E6 (EPD) waveform tuning Optical standard

Temperature (℃)		0~4	5~9	10~14	15~19	20~24	25~29	30~34	35~39	40~44	45~50	ΔE 2000
White state	TYP L*	66.5	66.5	66.5	66.5	66.5	66.5	66.5	66.5	66	65	≤5
	TYP a*	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	
	TYP b*	-2	-2	-2	-2	-2	-2	-2	-2	-2	-2	
Dark State	TYP L*	10	10	10	10	10	10	10	10	10	10	≤5
	TYP a*	10	10	10	10	10	10	10	10	10	10	
	TYP b*	-13	-12	-12	-12	-12	-12	-12	-12	-12	-12	
Red State	TYP L*	25	25	25	25	26	26	25	25	25	25	≤5
	TYP a*	39	40	40	40	41	41	40	40	40	40	
	TYP b*	29	30	30	30	30	30	30	30	30	30	
Yellow state	TYP L*	65	65	65	65	65	65	65	65	65	65	≤5
	TYP a*	-12	-12	-12	-12	-12	-12	-12	-12	-12	-12	
	TYP b*	67.5	67.5	67.5	67.5	67.5	67.5	67.5	67.5	67.5	67.5	
Blue State	TYP L*	30	30	30	30	30	30	30	30	30	30	≤5
	TYP a*	6	6	6	6	6	6	6	6	6	5	
	TYP b*	-40	-40	-40	-40	-39	-39	-38	-36	-35	-33	
Green State	TYP L*	34	34	34	34	35	35	35	35	33	33	≤7
	TYP a*	-19	-23	-23	-23	-24	-24	-24	-24	-24	-23	
	TYP b*	7	7	7	7	7	7	7	7	7	7	

WS : White state, KS : Black State,

Note 12-1 : Luminance meter : i - One Pro Spectrophotometer

Note 12-2

1. Optical measurement device and condition: I-ONE Pro3 plus with D65 light source 2° viewing angle for size above 5inch;
I-ONE Pro3 D65 light source 2° viewing angle for size below 5inch
2. The below character with background is recommended to use. Mura may occur if using non-recommended letter
eg: white/red/yellow/blue/green letter under black background
black/white/yellow/blue/green letter under red background
black/red/blue letter under white background
black/white/red letter under blue background
black/red letter under yellow background
black/yellow letter under green background



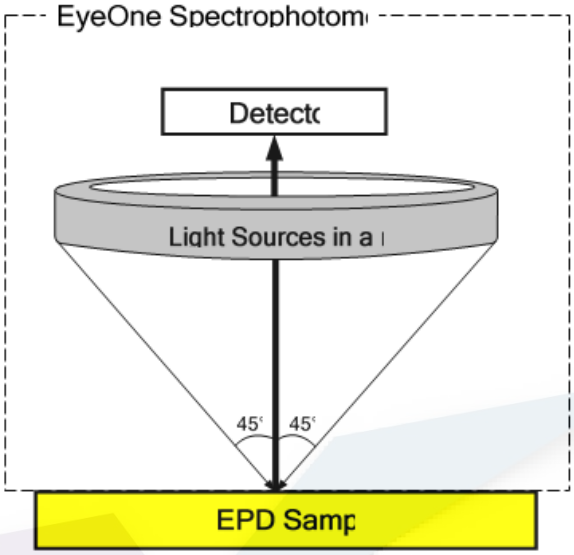
3. The judgement standard for ghosting is by 1A1B, and the Max 1A1B for green and black is 3; the othe color is 2
4. Mura judgement standard refers to the limited sample provided by seekink
5.Full temperature fonts do not control 1X1 pixels,, monitor fonts according to 2X2 pixels.

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11.3 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd)： R1: white reflectance Rd: dark reflectance

CR = Rl/Rd

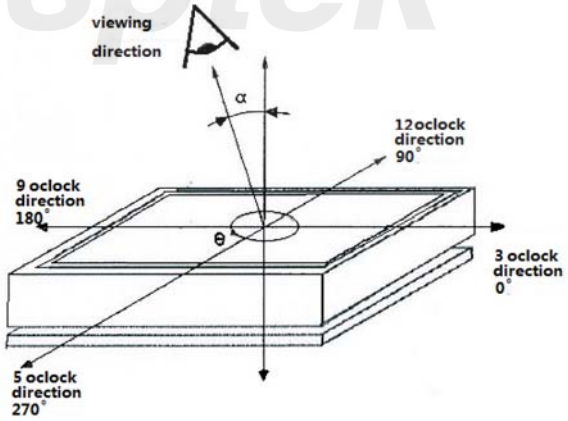


11.4 Reflection Ratio

The reflection ratio is expressed as:

R = Reflectance Factor white board x (L center / L white board)

L center is the luminance measured at center in a white area (R=G =B=1). L white board is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.



12. Handling Safety and Environmental Requirements

WARNING
The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap.

CAUTION
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.
Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

Mounting Precautions
(1) It`s recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.
(2) It`s recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.
(3) You should adopt radiation structure to satisfy the temperature specification.
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

Data sheet status	
Product specification	The data sheet contains final product specifications.

Limiting values
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

Application information
Where application information is given, it is advisory and dose not form part of the specification.

Product Environmental certification
ROHS

REMARK
All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.

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13. Reliability test

13.1 Reliability test items

	TEST	CONDITION	REMARK
1	High-Temperature Operation	T=40℃，RH=35%RH，For 240Hrs	
2	Low-Temperature Operation	T = 0℃ for 240 hrs	
3	High-Temperature Storage	T=50℃ RH=35%RH For 240Hrs	Test in white pattern
4	Low-Temperature Storage	T = -25℃ for 240 hrs	Test in white pattern
5	High Temperature, High-Humidity Operation	T=40℃，RH=90%RH，For 168Hrs	
6	High Temperature, High-Humidity Storage	T=50℃，RH=90%RH，For 240Hrs	Test in white pattern
7	Temperature Cycle	-25℃(30min)~60℃(30min)，100 Cycle	Test in white pattern
8	Package Vibration	1.04G,Frequency : 20~200Hz Direction : X,Y,Z Duration: 30 minutes in each direction	Full packed for shipment
9	Package Drop Impact	Drop from height of 100 cm on Concrete surface Drop sequence:1 corner, 3edges, 6face One drop for each.	Full packed for shipment
10	UV exposure Resistance	765 W/㎡ for 168hrs,40℃	
11	Electrostatic Discharge(no-operation)	（Machine model）:+/-200v，0Ω,200PF	

Actual EMC level to be measured on customer application.

Note1：Stay white pattern for storage and non-operation test.

Note2：Operation is black/white pattern , hold time is 150S.

Note3：Keep testing after 2 hours placing at 20℃-25℃.

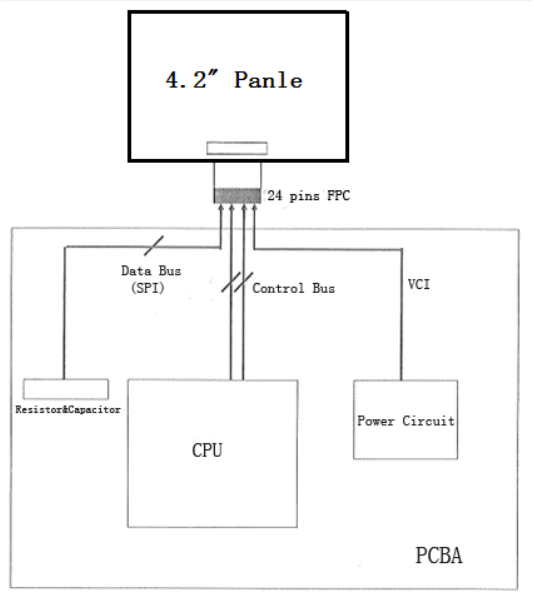
13.2 Product life time

The EPD Module is designed for a 3year life-time with 25 ℃/60%RH operation assumption. Reliability estimation testing with accelerated life-time theory would be demonstrated to provide confidence of EPD lifetime.

13.3 Product warranty

Warranty conditions have to be negotiated between Xingtai and individual customers. Xingtai provides 12+1(one month delivery time) months warranty for all products which are purchased from Xingtai.

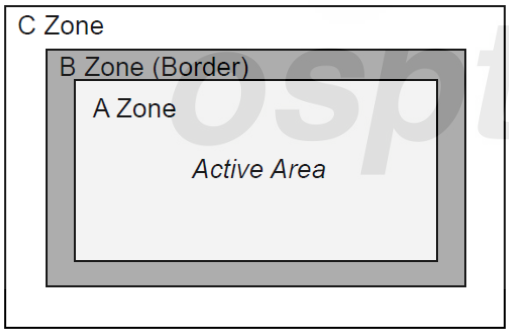
14. Block Diagram



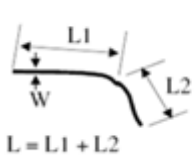
15. Shipment inspection specification

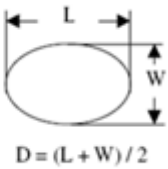
15.1 Zone Definition

A Zone: Active Area
B Zone: Border Area
C Zone: From B Zone edge to panel edge

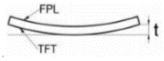


15.2 Line/Spot defect size





15.3 Point and line standard

<table>
<tr><td colspan="7">Shipment Inspection Standard</td></tr>
<tr><td colspan="7">Equipment：Electrical test fixture, Point gauge</td></tr>
<tr><td>Outline dimension</td><td>91.00(H)× 77.00(V) × 0.9(D)</td><td>Unit： ㎜</td><td>Part-A</td><td>Active area</td><td>Part-B</td><td>Border area</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td rowspan="2">Environment</td><td>Temperature</td><td>Humidity</td><td>illuminance</td><td>Distance</td><td>Time</td><td>Angle</td></tr>
<tr><td>19℃～25℃</td><td>55±5%RH</td><td>800~1300Lux</td><td>300 ㎜</td><td>35Sec</td><td>45°</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td>Defect type</td><td>Inspection method</td><td colspan="2">Standard</td><td colspan="2">Part-A</td><td>Part-B</td></tr>
<tr><td rowspan="3">Spot</td><td rowspan="3">Electric Display</td><td colspan="2">D≤0.25 ㎜</td><td colspan="2">Ignore</td><td>Ignore</td></tr>
<tr><td colspan="2">0.25 ㎜＜D≤0.4 ㎜</td><td colspan="2">N≤4</td><td>Ignore</td></tr>
<tr><td colspan="2">D＞0.4 ㎜</td><td colspan="2">Not Allow</td><td>Ignore</td></tr>
<tr><td>Display malfunction</td><td>Electric Display</td><td colspan="2">Not Allow</td><td colspan="2">Not Allow</td><td>Ignore</td></tr>
<tr><td>Display error</td><td>Electric Display</td><td colspan="2">Not Allow</td><td colspan="2">Not Allow</td><td>Ignore</td></tr>
<tr><td rowspan="3">Scratch or line defect(include dirt)</td><td rowspan="3">Visual/Film card</td><td colspan="2">L≤2 ㎜,W≤0.2 ㎜</td><td colspan="2">Ignore</td><td>Ignore</td></tr>
<tr><td colspan="2">2.0mm＜L≤5.0mm,
0.2＜W≤0.3mm,</td><td colspan="2">N≤2</td><td>Ignore</td></tr>
<tr><td colspan="2">L＞5.0 ㎜,W＞0.3 ㎜</td><td colspan="2">Not Allow</td><td>Ignore</td></tr>
<tr><td rowspan="3">PS Bubble</td><td rowspan="3">Visual/Film card</td><td colspan="2">D≤0.2mm</td><td colspan="2">Ignore</td><td>Ignore</td></tr>
<tr><td colspan="2">0.2mm≤D≤0.35mm</td><td colspan="2">N≤4</td><td>Ignore</td></tr>
<tr><td colspan="2">D＞0.35 mm</td><td colspan="2">Not Allow</td><td>Ignore</td></tr>
<tr><td rowspan="2">Corner /Edge chipping</td><td rowspan="2">Visual/Film card</td><td colspan="5">X≤6mm,Y≤0.4mm, Do not affect the electrode circuit (Edge chipping)
X≤1mm,Y≤1mm, Do not affect the electrode circuit((Corner chipping)
Ignore</td></tr>
<tr><td colspan="5">  </td></tr>
<tr><td>TFT warping</td><td colspan="6">For 1.54~7.5inch, T≤2mm; For above 7.5inch, T≤3mm  </td></tr>
<tr><td rowspan="3">Remark</td><td colspan="6">1.Cannot be defect & failure cause by appearance defect;</td></tr>
<tr><td colspan="6">2.Cannot be larger size cause by appearance defect;</td></tr>
<tr><td colspan="6">L=long W=wide D=point size N=Defects NO</td></tr>
</table>

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15.4 Barcode



FF

ABBBBBBBBB

CCCDDDEEEE

What is the QR code after picture being scanned and read? It consist of 30 numbers and letters, here is explanation for it.

QR Code: ABBBBBBBBBFFCCCDDDEEEEGHIJKKLL

① A means Factory code, e.g.: H (HINK)
② BBBBBBBBB refers to product model name, which is in consistent with specification part number. E.g.: E0213A189 or E0266A120; when the total code is less than 9, it would be added suffix with **. e.g.: E027A45** or E0154A78*
③ FF means production workshop
④ CCC means date of production
⑤ DDD means production batch
⑥ EEEE means FPL lot
⑦ G means MP/Sample/Trial/Repair
⑧ HIJ:(H——TFT;I——PS;J——EC Glue)
⑨ KK means IC type
⑩ LL means serial number

16. Packing

Full carton: 12 pcs tray with products and 1 pcs empty tray.
Last carton: less 12pcs tray with products and 1 pcs empty tray. The packager will add the right amount of EPE to box so that the added EPE's height approximately close to the EPE height around the inside of the box.

